

1.FB PLC

1-10.POU's

1-10-1.Programs

1-10-1-2.DataBase

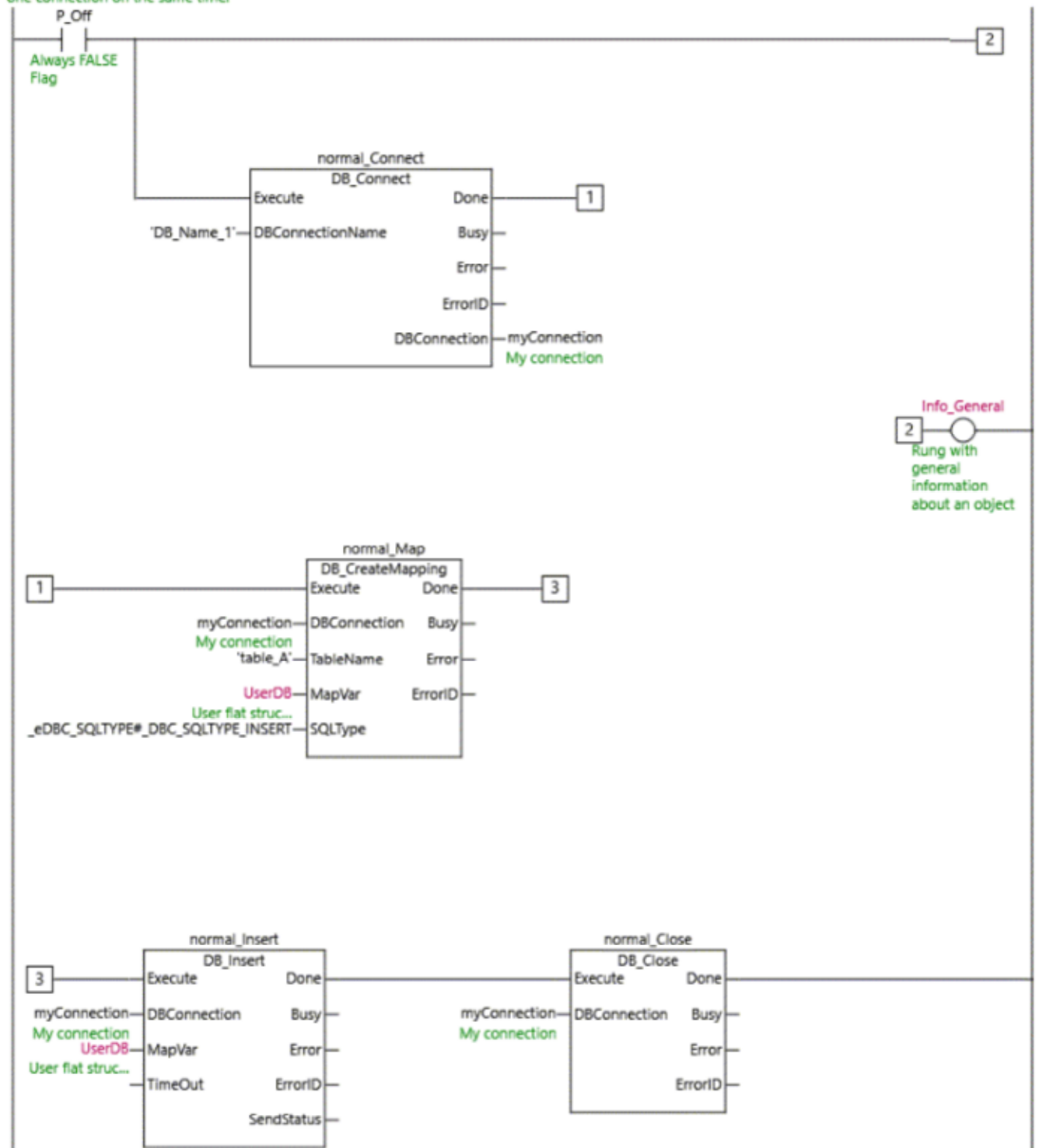
1-10-1-2-1.Variables

Name	Data Type	Initial Value	AT	Retain	Constant	Comment
VAR						
config_DB_1	OBJ\VB\VB_Manager			False	False	
normal_Connect	DB_Connect			False	False	
myConnection	DWORD			False	False	My connection
normal_Map	DB_CreateMapping			False	False	
normal_Insert	DB_Insert			False	False	
normal_Close	DB_Close			False	False	
config_DB_2	OBJ\VB\VB_Manager			False	False	
test_1	BOOL			False	False	
connect_A	OBJ\VB\VB_Connect_WithMgr			False	False	
test_A	BOOL			False	False	
connection_A	BOOL			False	False	I'm connected for "Test A"
done_A	BOOL			False	False	"Test A" is done
close_A	OBJ\VB\VB_Close_WithMgr			False	False	
test_B	BOOL			False	False	
A_Map	DB_CreateMapping			False	False	
map_A	DB_CreateMapping			False	False	
insert_A	DB_Insert			False	False	
connect_B	OBJ\VB\VB_Connect_WithMgr			False	False	
map_B	DB_CreateMapping			False	False	
insert_B	DB_Insert			False	False	
Close_B	OBJ\VB\VB_Close_WithMgr			False	False	
test_C	BOOL			False	False	
connect_C	OBJ\VB\VB_Connect_WithMgr			False	False	
map_C	DB_CreateMapping			False	False	
insert_C	DB_Insert			False	False	
Close_C	OBJ\VB\VB_Close_WithMgr			False	False	
test_All	BOOL			False	False	
VAR_EXTERNAL						
DB_Machine	OBJ\VB\BASE				False	
Info_General	BOOL				False	
UserDB	FlatDB\Users				False	
Info_Struct	BOOL				False	
DB_General	OBJ\VB\BASE				False	
Info_ToDo	BOOL				False	

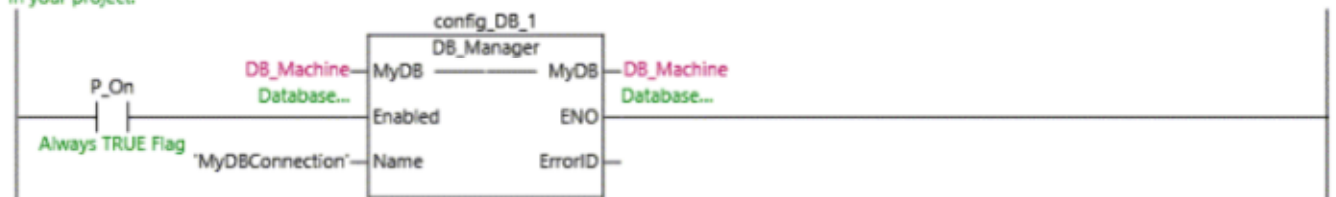
1-10-1-2-2.Config

- 0 Normally, the standard way open the communication and do something will be with like this example.

It's working, but will be complex if you have more than one communication request! You need to implement something to only have one connection on the same time.



- 1 This object will manage this problem for you! You need to instantiate only one "DB_Manager" by "DB Connection Setting" in your project.



- 2 The "DB_Manager" contains one "DB_Connect" function block, and must be the only one!

Every connection request will be passed through the structure.

Info_Struct

Rung with
information on
object structure

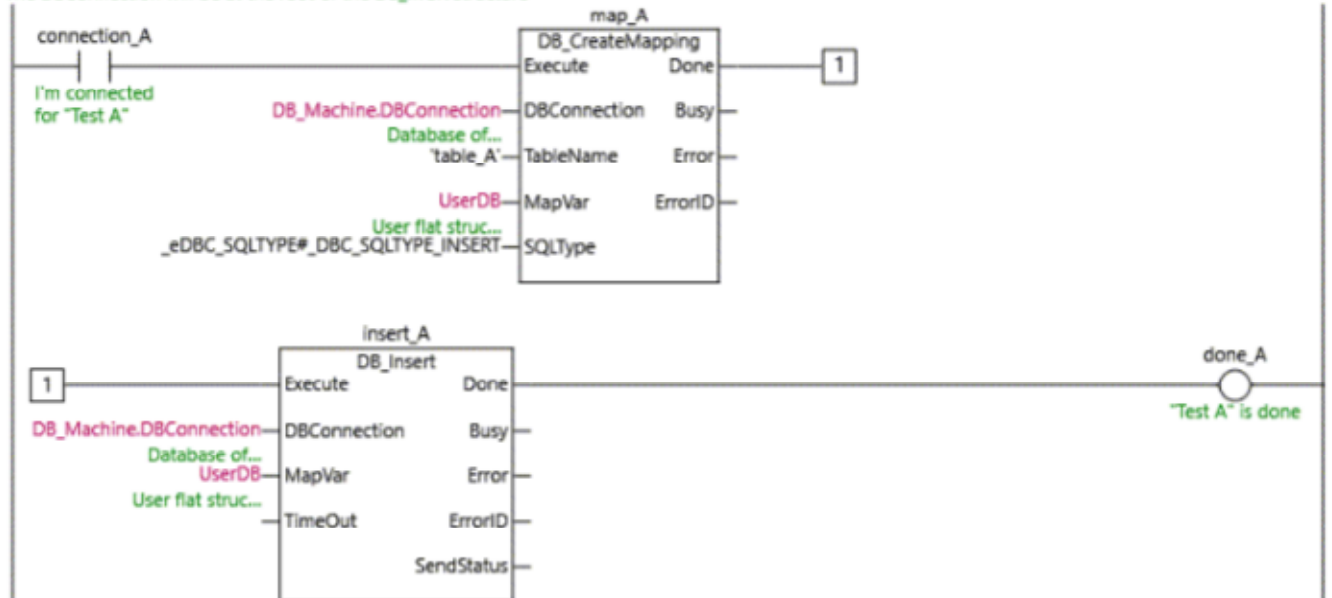
1-10-1-2-3.Exemple

- 0 The way to do something will be like this:
- The "Connected" output will rise when the connection is done.
 - It will stay at TRUE during the time of the connection (and still be TRUE if the "Request" input fail)



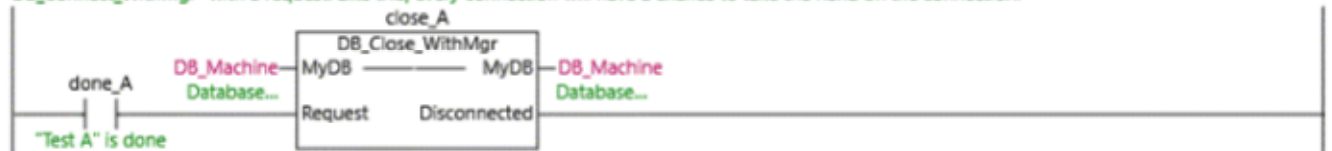
- 1 After, you can do your operation.

The DBConnection will be at the root of the DB_MGR structure



- 2 When your operation is finished, you can close the communication with this FB.

The order will be send to the manager, and return to this FB. The connection will be ready for the next instantiated "DB_Connect_WithMgr" with a request. Like this, every connection will have a chance to take the hand on the connection.

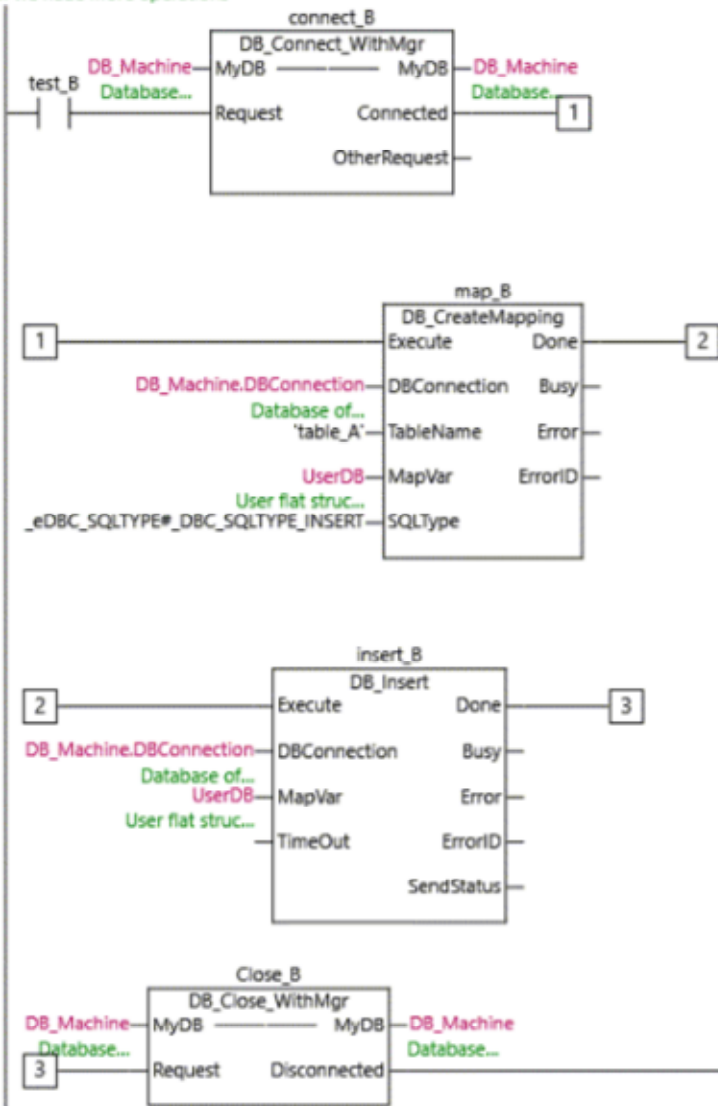


- 3 An open connect will stuck every other request. You must manager in you code to release the connection!
The output "OtherRequest" will informe you if another instance need the communication.

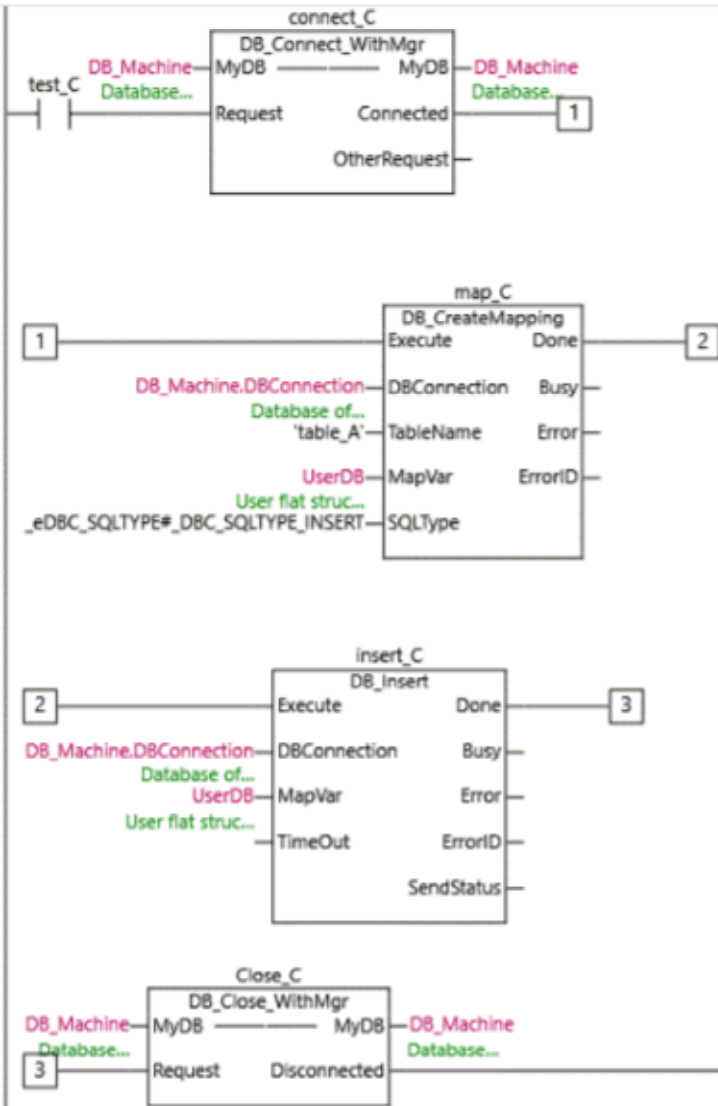
For exemple, if you have a batch to datas to send, you can stop if you see an external request



4 If we hade more operations

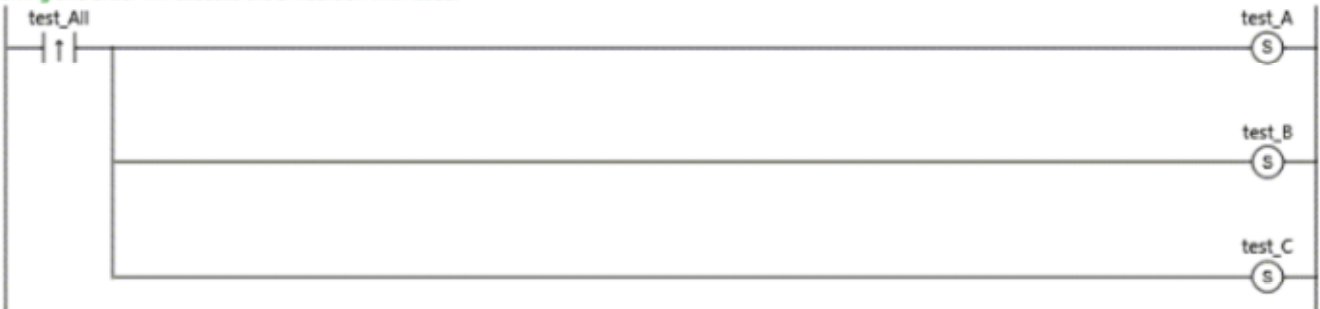


5



6

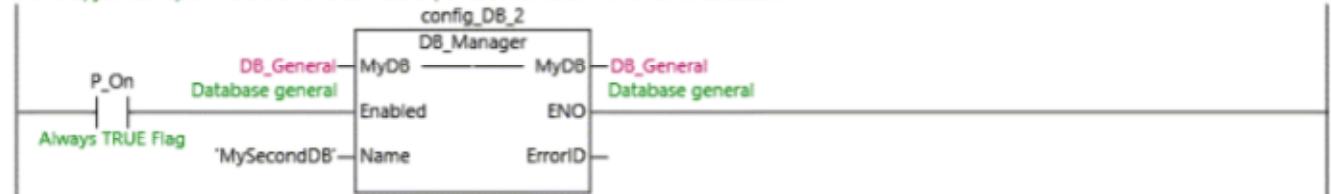
Giving this order will execute the 3 insertion with issue!



1-10-1-2-4SecondDB

0 You can create more than one connection.

With this, you can open more than one connection, or communicate with an other database.



1 Warning:
In some case (Old DB version, small PLC), it's not possible to have more than one open connection!

Having a seconde "DB_Manager" will make a connection fault!

The good way will be creating a new configuration "DB_Manager_Slave" with a seconde DB object, and the request will be for the master DB object

Info_ToDo
To do...

1-10-3.Function Blocks

1-10-3-9.DB Manager (OBJ\DB)

1-10-3-9-1.Variables

Name	Data Type	Edge	Initial Value	AT	Retain	Constant	Comment
VAR							
connectDB	DB_Connect				False	False	
closeDB	DB_Close				False	False	
getConnection	DB_GetConnectionStatus				False	False	
getService	DB_GetServiceStatus				False	False	
waitDeco	TON				False	False	
waitAutoReset	TON				False	False	
waitConnReset	TON				False	False	
orderConnection	BOOL				False	False	Connection order
Started	BOOL				False	False	DB service is running
initializing	BOOL				False	False	Initialization is running
error	BOOL				False	False	Error of connection
resetConn	BOOL				False	False	Reset comm
inFault	BOOL				False	False	Currently in fault
VAR INPUTOUTPUT							
ENO	BOOL	No Edge			False	False	
MyDB	OBJ\DATABASE	No Edge			False	False	DB
Enabled	BOOL	No Edge			False	False	Enable DB
Name	STRING[32]	No Edge			False	False	Name of the connection (In Sysmac - DB Connection settings)
ErrorID	WORD	No Edge			False	False	Connection error ID
VAR EXTERNAL							
T_Is	BOOL					False	

1-10-3-9-2.LadderBody

- 0
- S06V00R00 - 2026.05.08 - PC

- Use namespace
- S01V03R00 - 2024.06.26 - PCio

- Try to avoid a fault log at startup.

- Improve fault management

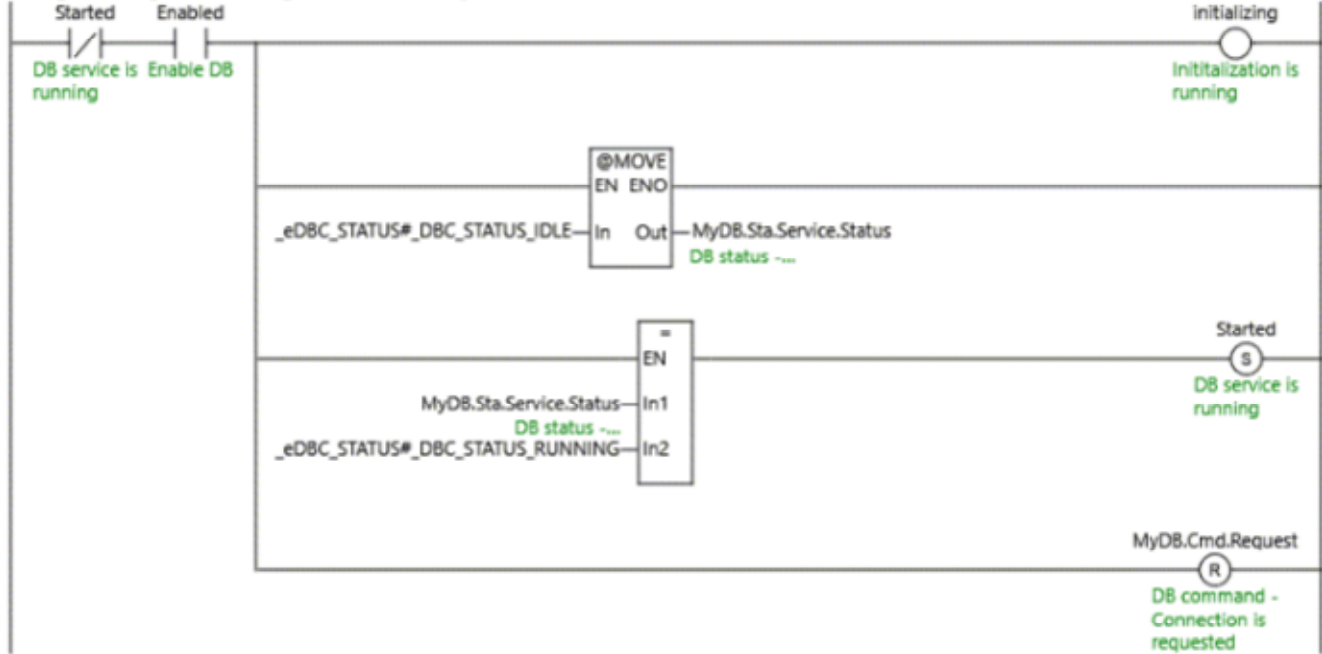
- Taking out logging fault
- S01V02R00 - 2024.06.19 - PC

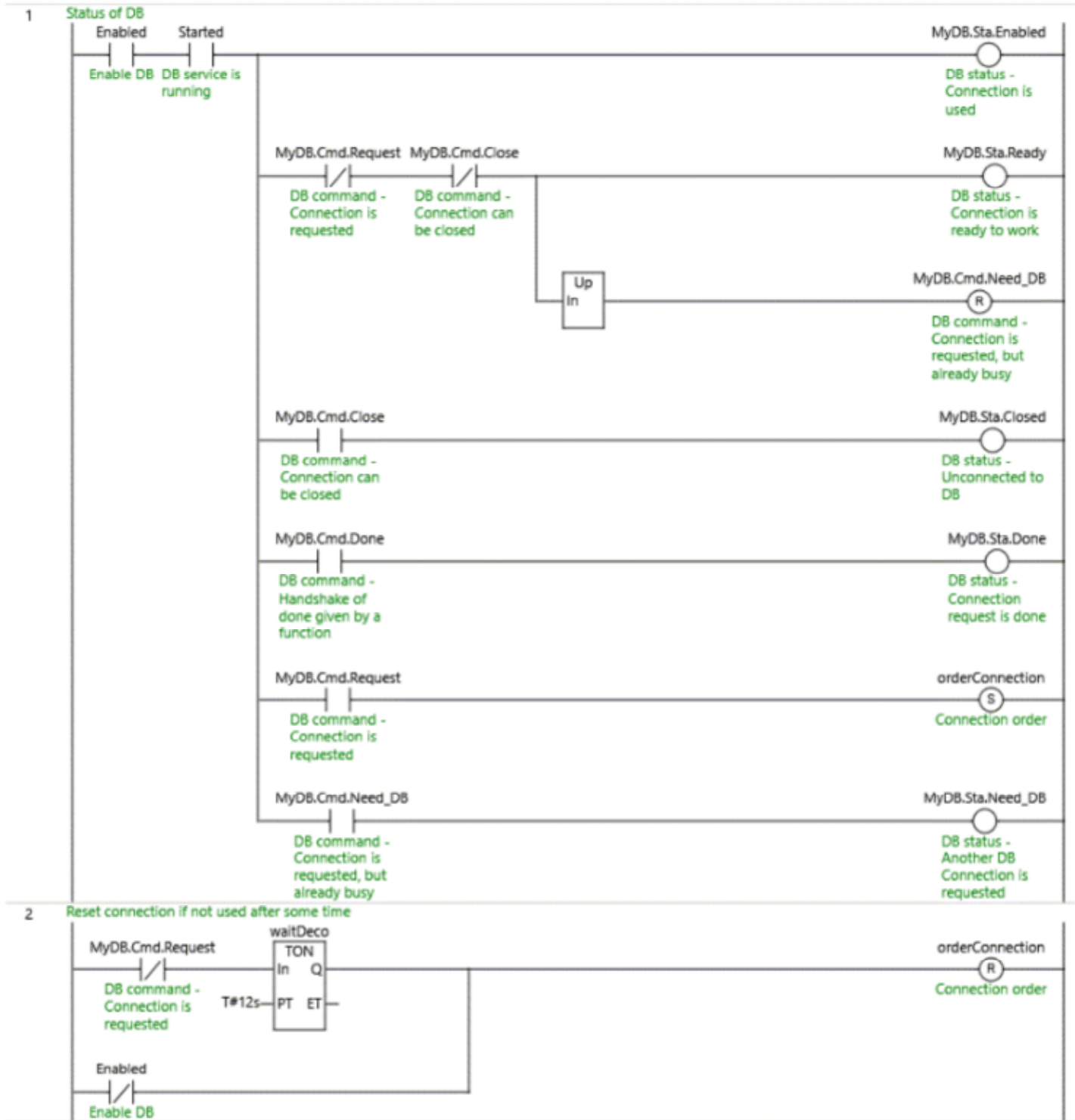
- Starting error management and logging
- S01V01R00 - 2024.06.11 - PC

- Add status and check before connection
- S01V00R00 - 2024.05.14 - PC

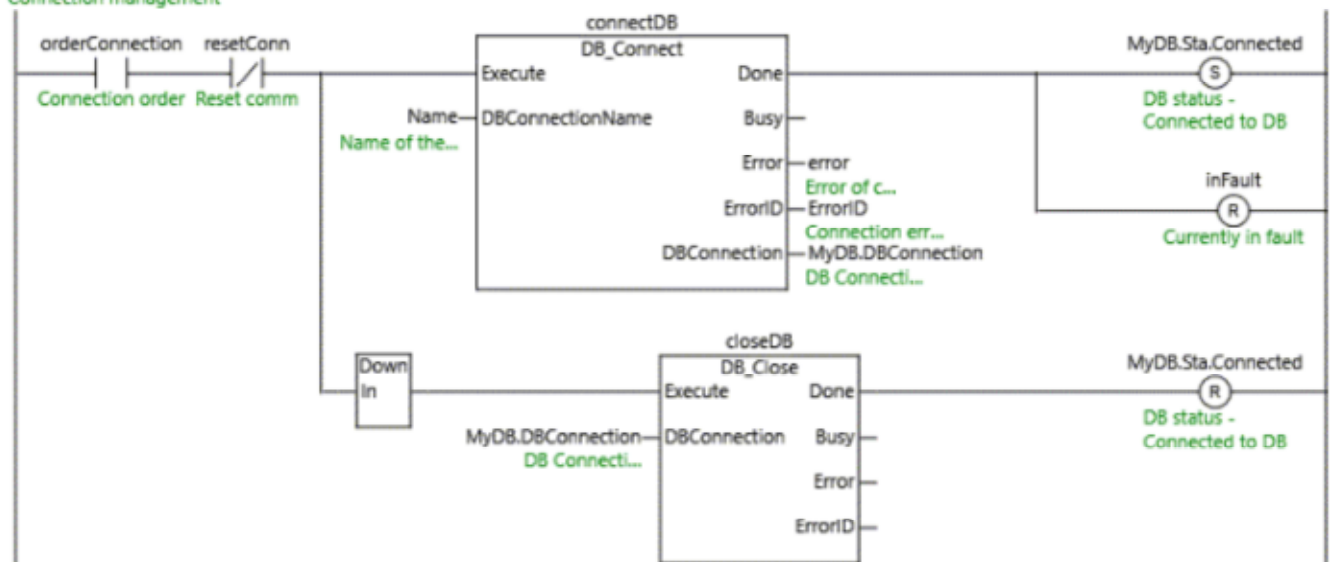
- First version

Starting DB manager and checking if DB service is ready to work

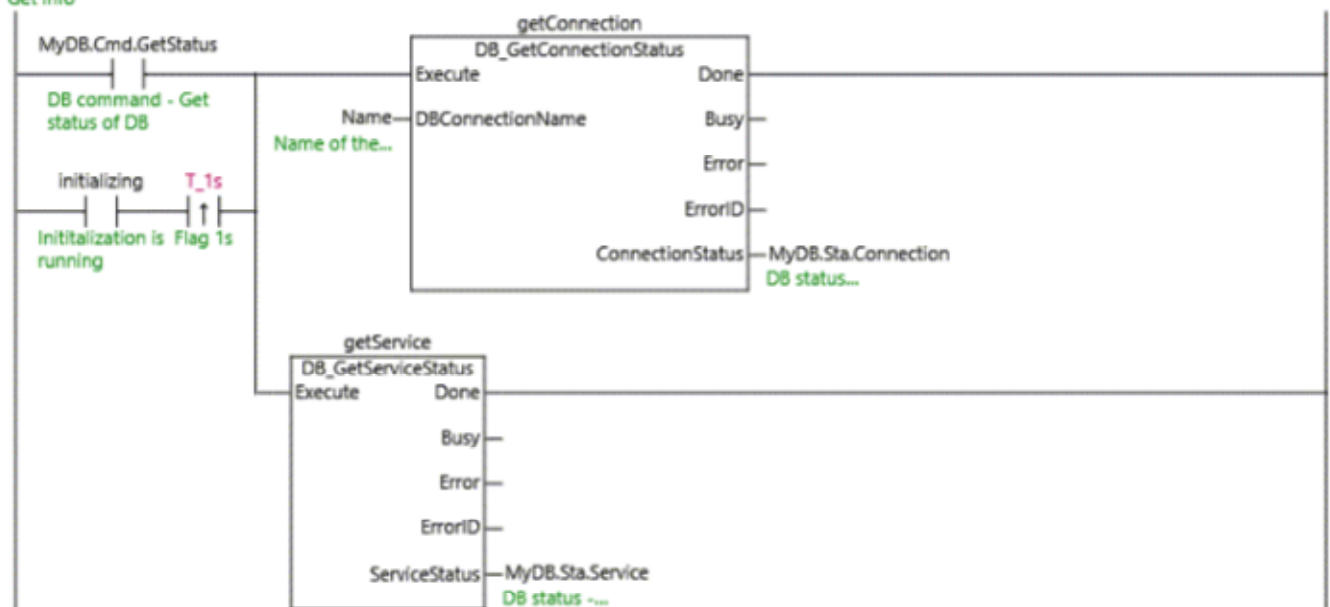




3 Connection management

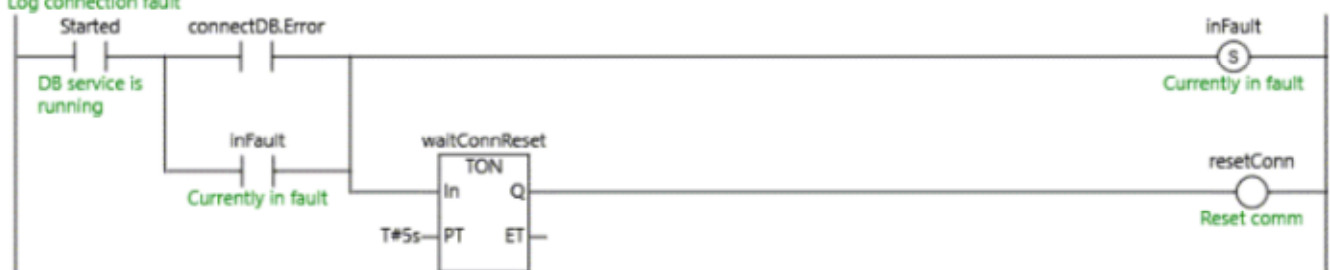


4 Get info

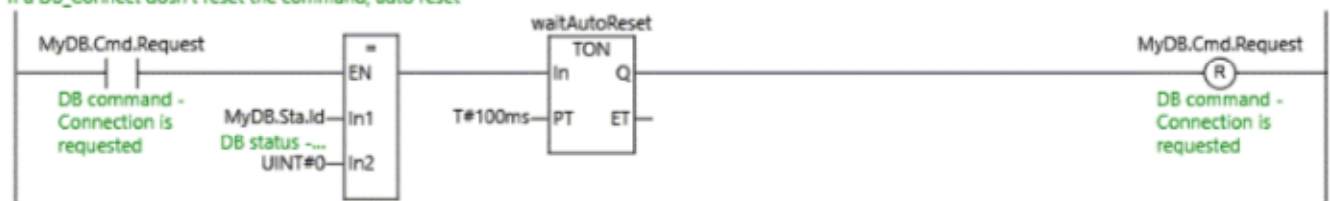


5 Error management

Log connection fault



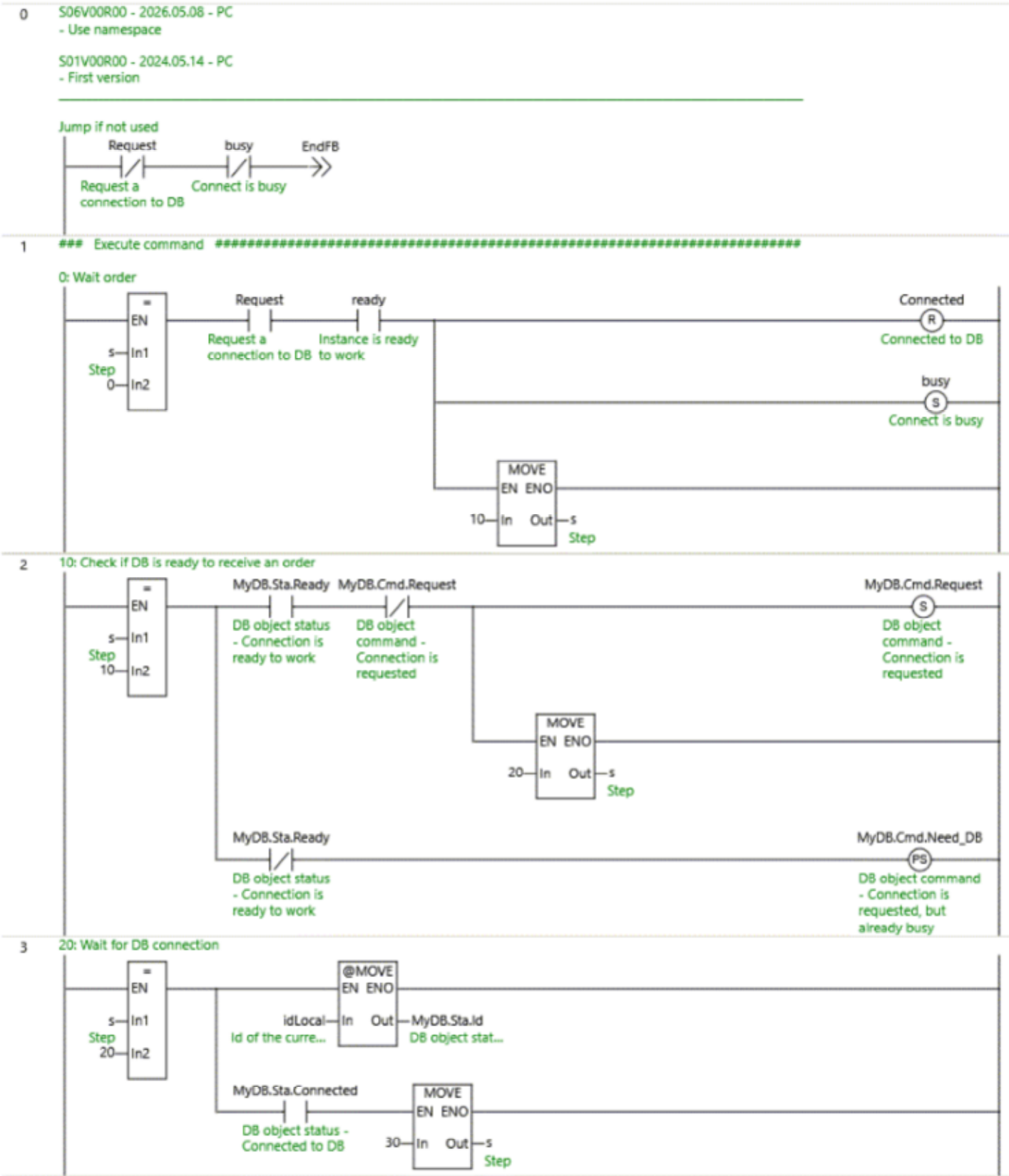
6 If a DB_Connect doesn't reset the command, auto reset



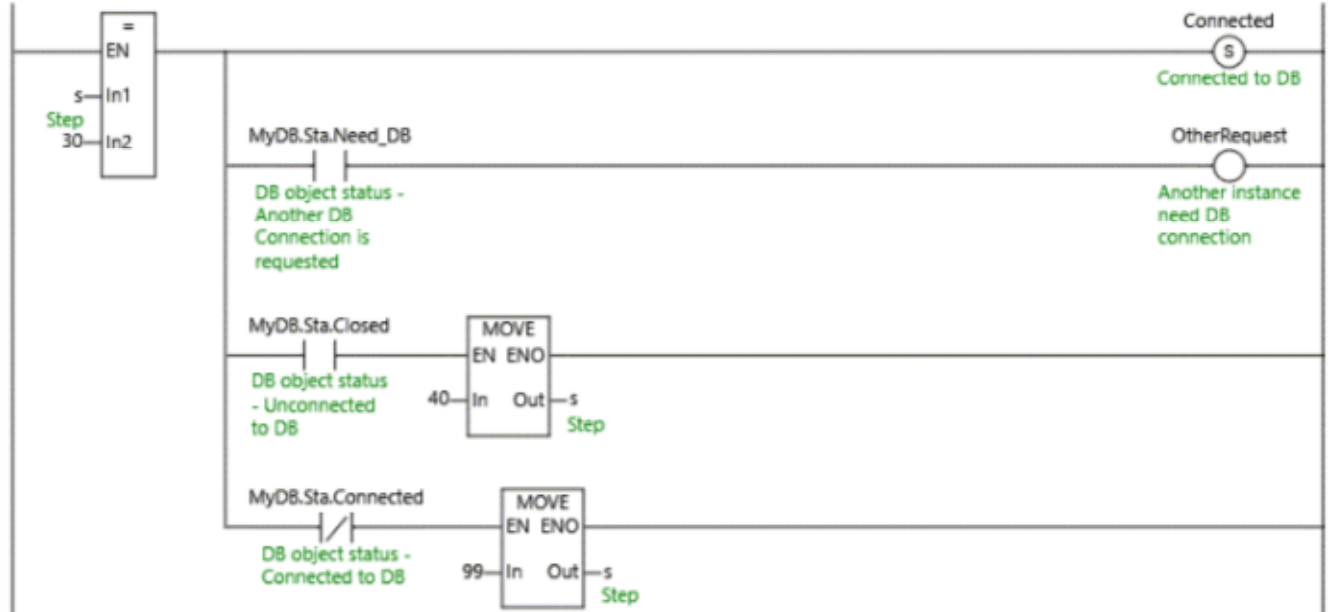
1-10-3-11.DB Connect WithMgr (OBJ\DB)**1-10-3-11-1.Variables**

Name	Data Type	Edge	Initial Value	AT	Retain	Constant	Comment
VAR							
idLocal	UINT				False	False	Id of the current instance
ready	BOOL				False	False	Instance is ready to work
s	INT				False	False	Step
busy	BOOL				False	False	Connect is busy
VAR INPUTOUTPUT							
MyDB	OBJ\DATABASE	No Edge			False	False	DB object
Request	BOOL	No Edge			False	False	Request a connection to DB
Connected	BOOL	No Edge			False	False	Connected to DB
OtherRequest	BOOL	No Edge			False	False	Another instance need DB connection

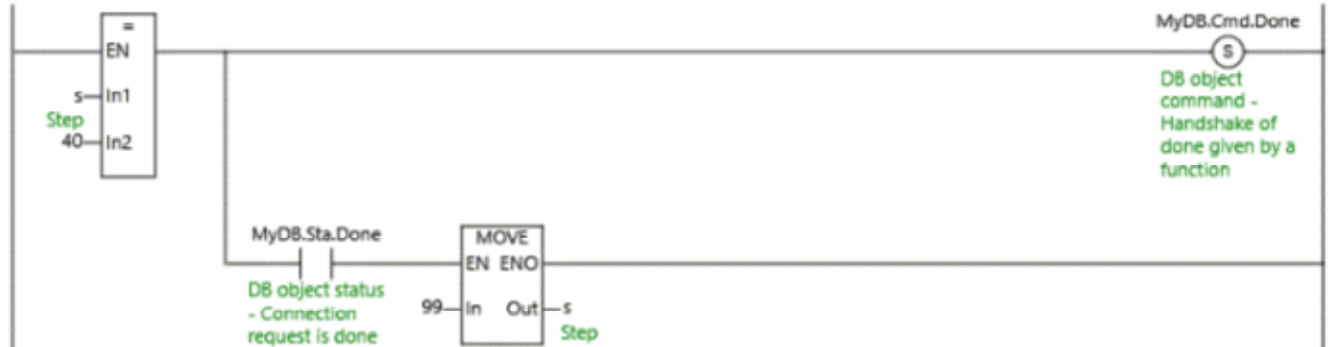
1-10-3-11-2.LadderBody



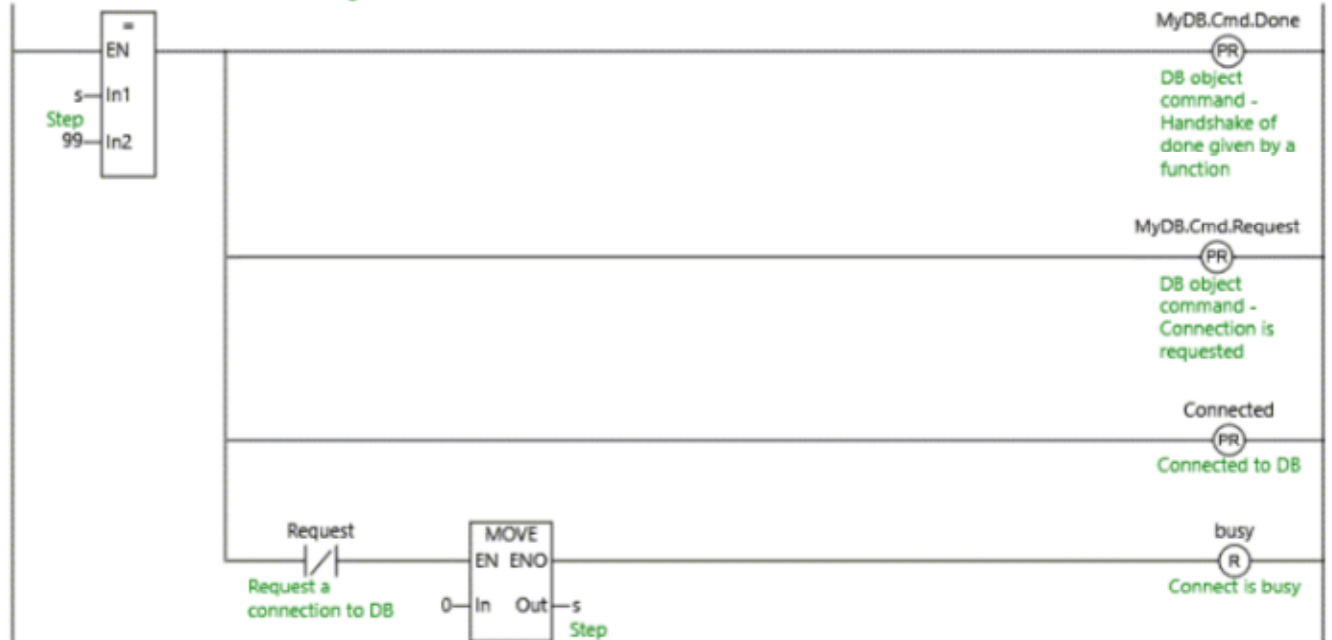
4 30: Connected to DB - Wait for an unconnection order



5 40: Handshake DB for done status

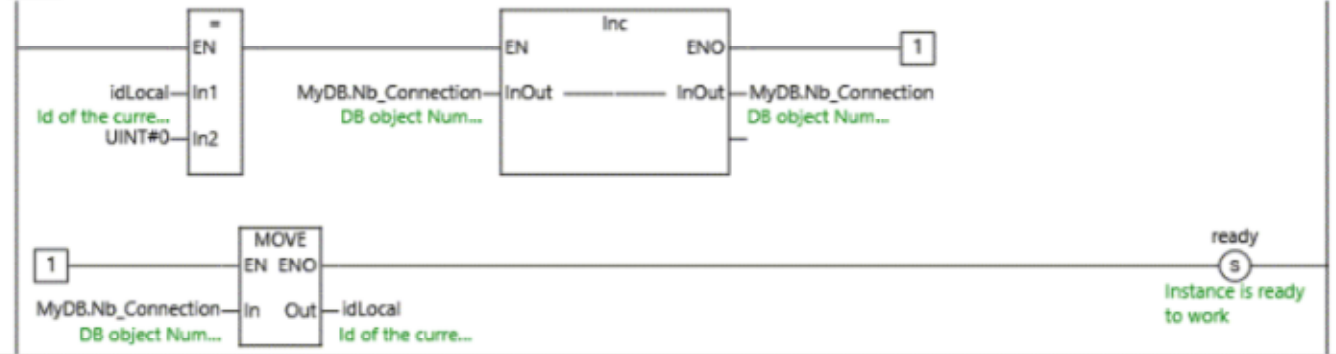


6 99: Reset done status to handshake config bloc



7 At startup, define the local id of instance

EndFB

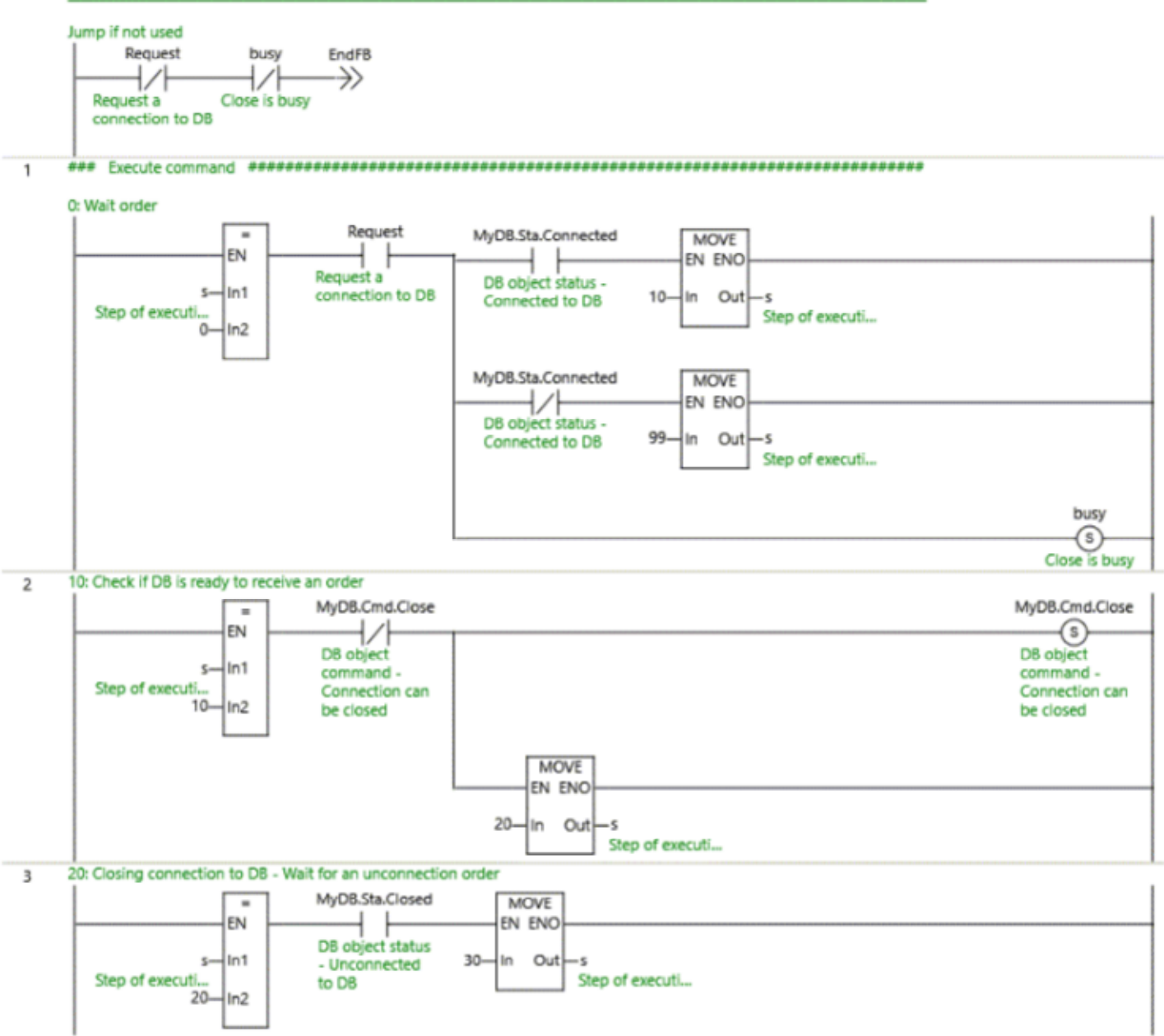


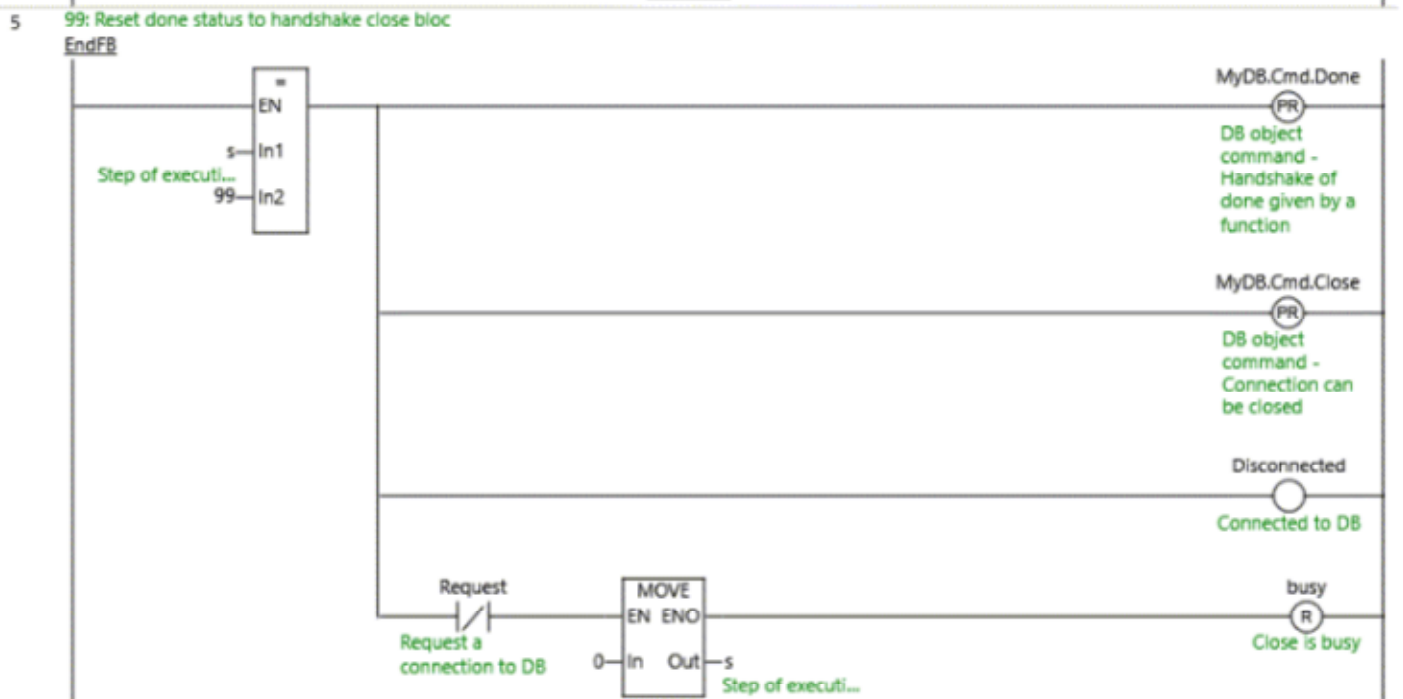
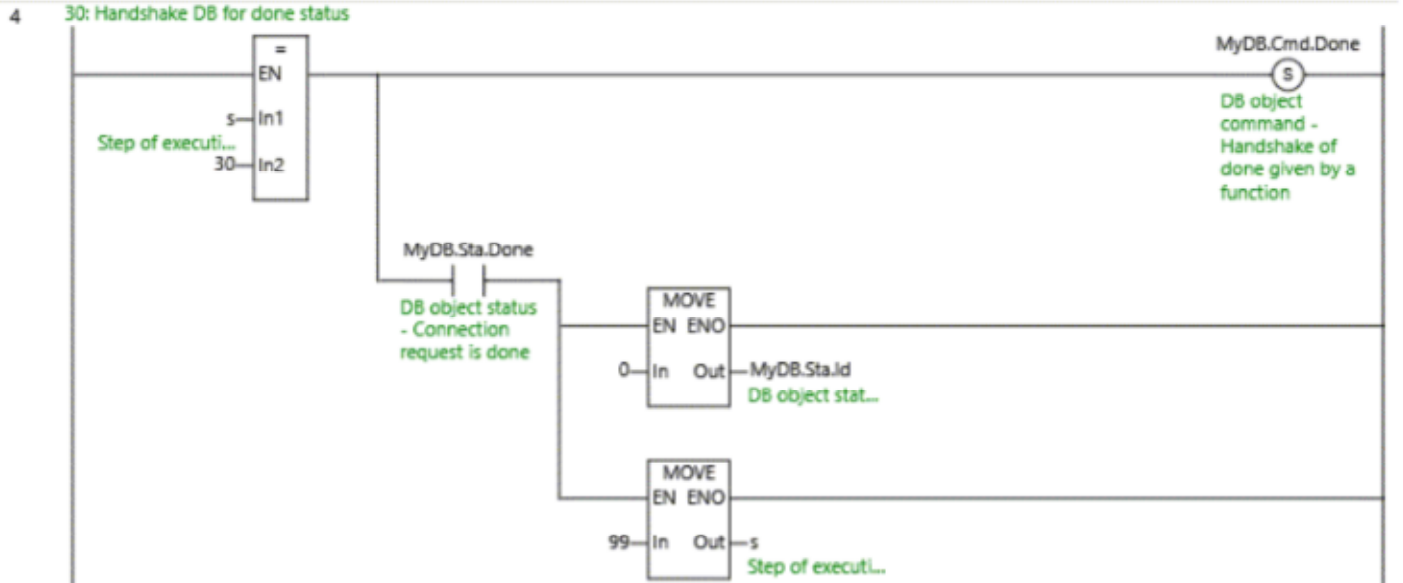
1-10-3-12.DB Close WithMgr (OBJ\DB)**1-10-3-12-1.Variables**

Name	Data Type	Edge	Initial Value	AT	Retain	Constant	Comment
VAR							
s	INT				False	False	Step of executing command
busy	BOOL				False	False	Close is busy
VAR INPUTOUTPUT							
MyDB	OBJ\DATABASE	No Edge			False	False	DB object
Request	BOOL	No Edge			False	False	Request a connection to DB
Disconnected	BOOL	No Edge			False	False	Connected to DB

1-10-3-12-2.LadderBody

0	S06V00R00 - 2026.05.08 - PC - Use namespace S01V00R00 - 2024.05.14 - PC - First version
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1-11.Data

1-11-5.OBJ\DB\Data Types

Name	Type		Offset Byte	Offset Bit	Comment
OBJ\DB\BASE	STRUCT	NJ			DB
Cmd	OBJ\DB\Command				command -
Sta	OBJ\DB>Status				status -
Msg	OBJ\DB\Message				message -
Nb_Connection	UINT				Number of object requesting DB
DBConnection	DWORD				Connection ID
OBJ\DB\Command	STRUCT	NJ			
Request	BOOL				Connection is requested
Need_DB	BOOL				Connection is requested, but already busy
Close	BOOL				Connection can be closed
Done	BOOL				Handshake of done given by a function
GetStatus	BOOL				Get status of DB
OBJ\DB>Status	STRUCT	NJ			
Enabled	BOOL				Connection is used
Ready	BOOL				Connection is ready to work
Connected	BOOL				Connected to DB
Closed	BOOL				Unconnected to DB
Done	BOOL				Connection request is done
Need_DB	BOOL				Another DB Connection is requested
Id	UINT				Instance ID of current connection
Connection	_sDBC_CONNECTION_STATUS				Connection status
Service	_sDBC_SERVICE_STATUS				Service status
OBJ\DB\Message	STRUCT	NJ			
TooMuchMgr	BOOL				More then one manager are instantiated
TooMuchDB	BOOL				More then one DB manager are instantiated
TooMuchCfg	BOOL				Too much config are used - You need to increase the array size